

Introduction to Big Data
Chapter 0 (Prerequisite)
Principal component analysis (PCA)

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January 4, 2021

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Nature of the data

The principal component analysis (PCA) is a method of exploratory multivariate data analysis which belongs to the family of factorial methods.

The data are presented in a table X where rows are considered as individuals and columns as quantitative variable. The size of the table is $n \times p$ where n is the number of individuals and p is the number of quantitative variables.

The application example deals with the 21 French regions except Corsica (the individuals or cases) characterized by several indicators (the variables).

Example

The considered variables are the following:

- POPUL : population of the region (in thousands of individuals)
- TACT : activity rate of the region (in percentage)
- SUPERF : surface of the region (in square kilometers)
- NBENTR : number of firms of the region
- NBBREV : number of patents taken out during the year
- CHOM : unemployment rate (in percentage)
- TELEPH : number of telephone lines in place in the region (in thousands)

Objectives of PCA

PCA has two principal objectives:

- *Summarize* the table X in a small number k of new variables uncorrelated between them and which will keep the maximum of the information contained in the p initial variables.

Intuitively, the new variables are obtained by “mixing together” the initial variables which are well correlated between each others.

The number k of these new variables is smaller if the correlations between the p initial variables are large.

As a by-product, PCA leads to a visualization of the correlations between the initial variables.

- *Interpret* the table X using the new variables and graphics such as scatterplots.

PCA allows in particular to detect outliers or to find groups of individuals having the same behavior with respect to the considered variables.

PCA principle

Information definition

The aim of “keeping the maximum of information contained in a set of data” implies that a concept of information is mathematically defined. This information is based on the variability of the data and is measured by the variance.

Definition :

- The information brought by a quantitative variable X^j is the variance of X^j .
- The information brought by a set of data $X = [X^1, X^2, \dots, X^p]$ is the sum of the variances of the variables of X . It is called *inertia* of X and is denoted I_X . We have :

$$I_X = \sum_{j=1}^p \text{var}(X^j).$$

PCA principle

Standardized variables

$$x^j = \frac{X^j - \overline{X^j}}{\sigma_{X^j}}$$

So $\overline{x^j} = 0$ and $\text{var}(x^j) = 1$.

Therefore, standardized variables bring the same information (equal to 1).
The standardized table is denoted by x .

Remarks

- $r(x^k, x^j) = r(X^k, X^j)$.
- In this course, PCA is performed on standardized variables. It is a particular case of PCA sometimes called *standardized* PCA.

PCA principle

Principal components (factors) definition

The **first principal component** c^1 is defined as a new variable, linear combination of the variables $x^1, x^2, \dots, x^p$ as follows :

$$c^1 = \alpha_1 x^1 + \alpha_2 x^2 + \dots + \alpha_p x^p \text{ avec } \sum_{j=1}^p \alpha_j^2 = 1 \quad (1)$$

and such that the information brought by c^1 is maximal.

In other words, the coefficients $\alpha_1, \alpha_2, \dots, \alpha_p$ are searched so that c^1 is of maximal variance.

The **second principal component** c^2 is defined as a new variable *uncorrelated with* c^1 , a linear combination of the variables $x^j, j = 1, \dots, p$ and of maximal variance.

The **third principal component** c^3 is *uncorrelated* with c^1 and c^2 , a linear combination of the x^j 's and of maximal variance.

...

The **p th principal component** c^p is *uncorrelated* with $c^1, c^2, \dots, c^{p-1}$, a linear combination of the x^j 's and of maximal variance.

p uncorrelated factors are so defined and are grouped in a table denoted $C = [c^1, c^2, \dots, c^p]$.

According to the previous definition, $\text{var}(c^1) \geq \text{var}(c^2) \geq \dots \geq \text{var}(c^p)$.

Indeed, each factor is defined according to a variance maximization criterion but with one more constraint for c^2 than for c^1 (correlation coefficient equal to 0 between c^2 and c^1), for c^3 than for c^2 (correlation coefficient equal to 0 between c^3 and c^2), ..., for c^{p-1} than for c^p (correlation coefficient equal to 0 between c^p and c^{p-1}).

Moreover, it is shown that the information brought by the table x is completely restored in the table C . In other words, the inertia of C is equal to the inertia x :

$$I_x = \sum_{j=1}^p \text{var}(x^j) = p = I_C = \sum_{j=1}^p \text{var}(c^j).$$

However, whereas each column of x brings the same information (equal to 1), the columns of table C bring an information which decreases with the number of the column.

The first objective of PCA is then reached: table x can be summarized by a table containing less columns if the last components bring little information (i.e. they have a small variance).

Remark : PCA can be introduced by other criteria than variance maximization. The geometric approach is notably often presented.

Principal components: computation and properties

Computing the factors

- Computing the correlation matrix:

$$R = \frac{1}{n} x' x.$$

- Computing the eigenvalues and eigenvectors: The eigenvalues and the eigenvectors of the correlation matrix R are denoted by $\lambda_j \in \mathbb{R}$ and the $v_j \in \mathbb{R}^p$ for $j = 1, \dots, p$ and are such that

$$R v_j = \lambda_j v_j.$$

The λ_j are decreasingly sorted:

$$\lambda_1 > \lambda_2 > \dots > \lambda_p$$

$$c^j = x v_j$$

Principal components: computation and properties

Properties of the principal components

Proposition

c^j is a linear combination of the initial variables.

Proposition

- $\overline{c^j} = 0$,
- $\text{var}(c^j) = \lambda_j$,
- $r(c^j, c^k) = 0$ for $j \neq k$,

Remark

$$I = p = \sum_{j=1}^p \lambda_j = \sum_{j=1}^p \text{var}(c^j)$$

Factor selection criteria

Explained variance criterion

One must choose a number k of principal components sufficient in order to summarize initial information without losing too much information.

Total information : $I = p$ and information brought by c^j : $\text{var}(c^j) = \lambda_j$,

proportion of inertia explained by c^1 : λ_1/p ,

proportion of inertia explained by c^1 and c^2 : $(\lambda_1 + \lambda_2)/p, \dots$

proportion of inertia explained by $c^1, c^2, \dots, c^k$: $\sum_{j=1}^k \lambda_j/p, \dots$

proportion of inertia explained by the p principal components. :

$\sum_{j=1}^p \lambda_j/p = p/p = 100\%$.

Criterion : k is chosen so that the proportion of explained variance is sufficiently large (at least 80 %).

Factor selection criteria

Kaiser criterion

The initial variables have a variance equal to 1 (standardized).

Criterion: select the principal components whose variance is > 1 because they provide more information than the initial variables ,

$$k = \text{number of } \lambda_j > 1.$$

Factor selection criteria

Scree test criterion

The differences between eigenvalues are examined :

$$\lambda_1 - \lambda_2, \lambda_2 - \lambda_3, \dots$$

In general, these differences decrease.

Criterion : retain the k principal components so that the difference $\lambda_k - \lambda_{k+1}$ is large and the differences $\lambda_j - \lambda_{j+1}$, $j = k + 1, \dots, p - 1$ are small.

Interpreting the factors

Presentation of the problem

Let us assume that, after using one of the previous criteria, k (small) principal components (or k dimensions or k factors) have been selected.

One of the difficulties of PCA (and of factorial analysis in general) is the interpretation of the principal components.

PCA leads to a reduction of the variables number (from p to k) but if the signification of the initial variables is known, it is not the case for the principal components.

Interpreting the factors

Interpretation of the coefficients of the linear combinations

$$c^j = x v_j, \forall j = 1, \dots, p$$
$$c^j = \sum_{k=1}^p v_j^k x^k$$

The “composition” of the c^j s’ is known and the important x^k variables are associated to large coefficients v_j^k (since they have the same variance). But this method is rarely used (the size of the coefficients is assessable with difficulty).

Interpreting the factors

Interpretation of the correlations between the components and the initial variables

- These correlations can be directly interpreted.
- Property :

$$r(c^j, x^k) = \sqrt{\lambda_j} v_j^k$$

- But in general, it is preferable to represent the correlations by considering the principal components 2 by 2 and to interpret them graphically (possible because k is small $\Rightarrow$ small number of possible correlation plots.)
- These correlations are obviously situated in a square $[-1, +1] \times [-1, +1]$ (correlation coefficients).
- In fact, it can be shown that the points are always situated in the circle centered at the origin and of rayon 1. This circle is often drawn because it is helpful for the interpretation.

In practice

- For each pair (or plane) $((c^1, c^2), (c^1, c^3), (c^2, c^3), \dots (c^{k-1}, c^k))$, draw the correlations.
- Draw the “correlations circle” (center at the origin and radius equal to one).
- Locate the strong correlations that is the points which are close to the circle. The correlations too much far from the circle are not interesting because because they do not intervene or little in the computing of the components and therefore are not useful for their interpretation.
- Interpret each component according to strong correlations (positive and negative).

Remark The quality of representation of a variable x^k on an axis c^j can be also measured by $r^2(x^k, c^j)$

Interpreting the individuals

Scatterplot of the individuals

We now have k new variables whose meaning is known.

In order to interpret the initial table, the individuals are represented on the new variables chosen 2 by 2 (possible since k is small). When 2 principal components are considered, the term of **principal plane** is used.

The obtained graphs are interpreted as any scatter plot taking into account the interpretation of the principal components. But, as for correlations, the individuals badly represented should not be interpreted.

Remark PCA can be also presented with a geometric approach as follows :

- A graph provides a synthetic view of the data.
- But scatterplots only possible for 2 variables (sometimes 3).
- Therefore, the observations are **projected** on a space $\mathbb{R}^k$ with k small, trying to loose the minimum of information.
- But the problem is that a projection involves a deformation of the distances (always shorter).
- Consequently, in order to interpret the graphs of individuals, the distances must be well preserved. As a matter of fact, points seemingly close, can be hardly far in the space on the other dimensions left aside by the graph.

Interpreting the individuals

Measure of the quality of representation of the individuals

In order to measure the quality of representation, we examine the **distance to the origin** of each individual.

- At the beginning, an individual x_i has a distance to the origin defined by :

$$d(x_i, O) = \sqrt{\sum_{j=1}^p (x_i^j)^2} = \sqrt{\sum_{j=1}^p (c_i^j)^2}$$

(also called the norm of the individual x_i : $d(x_i, O) = \|x_i\|$).

- After projection, the norm becomes :

- ▶ on a k -dimension space,

$$\|P_{X_i}\|_k = \sqrt{\sum_{j=1}^k (c_i^j)^2}.$$

- ▶ on a factor c^j : $\|P_{X_i}\|_1 = |c_i^j|$.

For each individual, the initial squared norm is compared to the squared norm after projection by computing the ratio of the two norms, that is for an axis c^j :

$$\frac{(c_i^j)^2}{\sum_{l=1}^p (x_i^l)^2} = \frac{(c_i^j)^2}{\sum_{l=1}^p (c_i^l)^2} = \cos^2 \theta_i^j.$$

Remark

$$\sum_{j=1}^p \frac{(c_i^j)^2}{\sum_{l=1}^p (c_i^l)^2} = 100\%.$$

“Size” effect

- When all the initial variables are positively correlated between them , the first principal component defines a size factor.
- A symmetric matrix (in our case the correlation matrix) having all its terms positive possesses a first eigenvector whose coordinates have the same sign.
- If they are chosen positive, the first principal component is then positively correlated with all the variables.

(Varimax) rotation

- One of the difficulties of PCA is the interpretation of the axes.
- But when there are numerous variables with average correlations, the interpretation is difficult.
- The role of rotation methods is to return the more clear-cut correlations, making the axes revolve which involves an easier reading.
- The VARIMAX rotation makes the axes turn preserving their orthogonality but the first factor is not any more the axis of larger variance.

The Varimax rotation aims of maximizing the variance of the correlations in each column of the table of the correlations between the principal components and the initial variables.

Conclusion

PCA is a statistical method which can be applied :

- to a table individuals / variables
- for p quantitative variables
- $p > 3$
- when some variables are well correlated between each others

Remark

If $R=Id$, then PCA is not useful.